

CENTER OF LEARNING EXCELLENCE · ABBVIE

ARCH

Knowledge Experience Platform (KEP)

Personalized Learning · Knowledge on Demand · Semantic Intelligence

Transforming how AbbVie employees discover, learn, and act on knowledge

THE KNOWLEDGE ACCESS PROBLEM

TODAY'S REALITY

- ✗ **Siloed Systems** Library, DMS, LMS, research repos — each requires separate login & search skills
- ✗ **Vocabulary Barrier** Same concept, 5 different terms across departments — discovery fails
- ✗ **Zero Personalization** Same content presentation for an MSL and a QA engineer
- ✗ **Learning ≠ Knowledge** Training completion ≠ access to relevant knowledge when it matters
- ✗ **No Value Signal** No way to know if learning investments are closing real knowledge gaps

THE ARCH VISION

- ✓ **Unified Knowledge Graph** All content sources federated through semantic ontology layer
- ✓ **Persona-Aware Vocabulary** Every query resolved in the employee's domain language
- ✓ **Adaptive Personalization** Role, JTBD, and proficiency drive content surfacing
- ✓ **Learning–Knowledge Fusion** Formal learning & knowledge discovery are one seamless experience
- ✓ **Value Intelligence** UX metrics continuously reveal value impact and identify value gaps

The gap between these two realities is bridged by a semantic knowledge architecture — not a better search engine.

ARCHITECTURE OVERVIEW

Six integrated layers from raw content to personalized knowledge experience

6 PERSONALIZED UX & METRICS LAYER
Conversational Interface · Value Measurement · Abstraction Boundary

5 DOMAIN AI AGENTS
Scientific · Regulatory · Learning Pathways · Library & Discovery

4 SEMANTIC FRAMING & PRECISION PROMPTS
Context Assembly · Agent Orchestration · Grounded Generation

3 NLP → SPARQL TRANSLATION
Intent Classification · Entity Resolution · Query Expansion

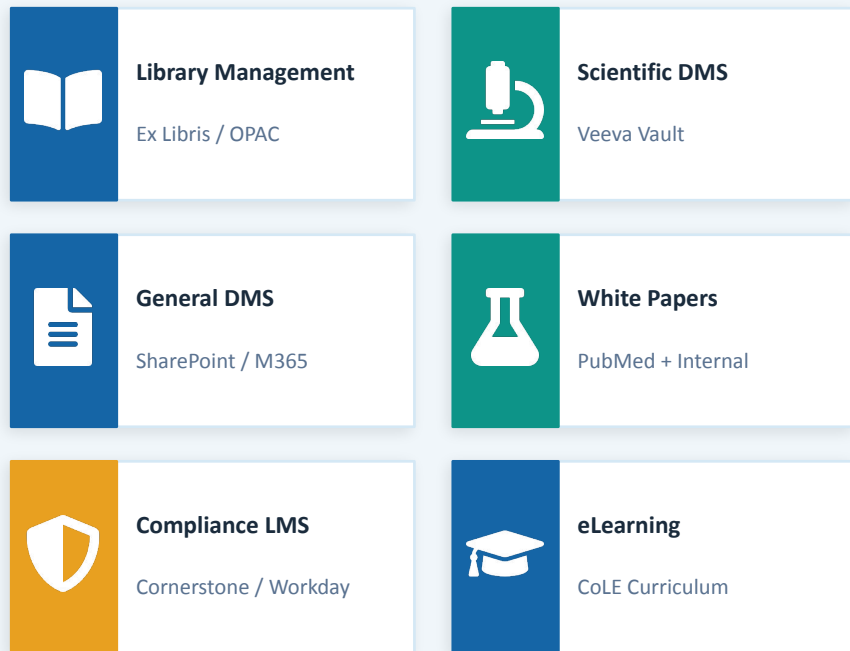
2 PERSONA ONTOLOGY & JOBS-TO-BE-DONE
Role Vocabulary · Proficiency Model · Adaptive Relevance

1 CONTENT FEDERATION & KNOWLEDGE GRAPH
Library · DMS · LMS · Research · White Papers · Compliance

↑ User Experience ↓

LAYERS 1 & 2: CONTENT FEDERATION & PERSONA INTELLIGENCE

Knowledge Sources Connected



Persona Ontology + JTBD Model

Role → Domain Vocabulary

MSL, Regulatory, QA, R&D — each mapped to terminology set

Job-to-be-Done Patterns

Query intent mapped to knowledge need clusters

Proficiency Level

Depth & register of content presentation auto-calibrated

Adaptive Relevance

Prior access patterns feed continuous relevance weighting

All content is semantically fingerprinted on ingestion — entities extracted, relationships mapped, vocabulary tagged, provenance recorded.

LAYER 3: ZERO-TRAINING NLP INTERFACE (→ SPARQL generated)

1

Natural Language Input

Employee types or speaks a question in their own words — no query syntax required. No system training. No controlled vocabulary.

2

Intent Classification

System resolves whether this is a learn, find, verify, compare, or comply need — routing to the right knowledge pipeline.

3

Persona-Calibrated Entity Resolution

"Safety profile" from an MSL maps to different ontology nodes than from a QA engineer. Disambiguation is automatic.

4

SPARQL Query Generation

Resolved entities + intent → structured SPARQL against the ARCH knowledge graph. Precision, not keyword matching.

5

Semantic Query Expansion

Ontology traversal includes related concepts the user didn't explicitly name but almost certainly needs.

Key Insight: This interface works on Day 1 for every employee. The persona model and semantic layer do the translation invisibly — employees simply describe what they need in their own professional language.

LAYERS 4 & 5: SEMANTIC FRAMING + DOMAIN AI AGENTS

LAYER 4: Semantic Context Builder

- 1 SPARQL Results In**
Retrieved knowledge graph subgraph relevant to the query
- 2 Agent Selection**
Identifies which domain-specific agents are appropriate for this query type
- 3 Context Assembly**
Builds structured grounding from ontology relationships — not just similar documents
- 4 Precision Prompt Engineering**
Constructs prompts with semantic context embedded — AI answers FROM the graph
- 5 Hallucination Prevention**
Semantic layer acts as truth constraint — agents can only respond from graph contents

LAYER 5: Domain AI Agents

- Scientific Knowledge Agent**
Tuned on AbbVie research, white papers, clinical data.
Answers with citation provenance.
- Regulatory & Compliance Agent**
Grounded in SOPs, regulatory submissions, compliance training requirements.
- Learning Pathways Agent**
Traverses curriculum ontology to recommend sequences — not just individual assets.
- Library & Discovery Agent**
Semantic-ranked full-text search across all document repositories.
- Agent Orchestrator**
Routes queries to the right combination of agents and synthesizes results coherently.

LAYER 6: THE UX AS STRATEGIC ABSTRACTION LAYER

The UX is not decoration — it is a deliberate architectural boundary with three strategic functions.

I Technology Abstraction



Employees interact with a stable, consistent experience. The underlying knowledge services — ontologies, agents, data sources — can evolve, expand, and be replaced without disrupting the user. Continuous improvement becomes invisible to the workforce.

II Value Measurement Engine



Every interaction is a signal. Query patterns, satisfaction scores, result acceptance, follow-up behavior, time-to-answer — captured, structured, and fed into a Value Intelligence layer that tracks ROI and identifies gaps in real time.

III Value Articulation Platform



Aggregated UX data tells the story of impact: knowledge gaps closed, training needs surfaced, content utilization, expert networks activated. This makes the invisible value of knowledge investment visible to CoLE leadership.

UX DESIGN: METRICS CAPTURE & VALUE SIGNALS

Personalized Librarian / Mentor Experience

Conversational Interface

Zero-training NLP — employees speak naturally

Semantic Summarization

AI-synthesized response calibrated to proficiency level

Provenance Transparency

Every claim cites its source with direct access link

Adaptive Guidance

"Based on your role & recent work, you may also need..."

Learning Path Integration

Knowledge gaps auto-link to formal learning curriculum

Metrics Captured at Every Interaction

Query Intent Distribution

Learn vs. Find vs. Verify — reveals knowledge access patterns

Result Acceptance Rate

Did the user engage with what was returned?

Content Utilization by Asset

Which sources are high-value vs. unused

Time-to-Answer

Tracks how quickly employees get what they need

Follow-up Depth

Did the answer satisfy or trigger further questions?

Persona Gap Patterns

What knowledge is consistently missing for a role?

Metrics are structured not just for reporting — they feed back into the persona model and relevance engine, creating a continuous improvement loop.

VALUE MEASUREMENT FRAMEWORK

Measuring what matters — from usage signals to organizational impact

TIER 1 Efficiency Value

- Time-to-Knowledge: avg. minutes to answer a domain question
- Search Deflection: queries resolved without human expert escalation
- Content Discovery Rate: % of relevant assets surfaced vs. found manually
- Onboarding Acceleration: time to productivity for new roles

TIER 2 Effectiveness Value

- Knowledge Application Rate: evidence that knowledge was acted on
- Learning Transfer Index: correlation between training & knowledge access
- Decision Confidence Score: self-reported certainty post-answer
- Compliance Readiness: gap between training completion & knowledge demonstrated

TIER 3 Strategic Value

- Knowledge Gap Map: systemic gaps by persona, domain, lifecycle stage
- Content Investment ROI: utilization vs. cost of assets by repository
- Expert Network Discovery: which people are queries escalating to?
- Innovation Signal: novel query patterns indicating emerging knowledge needs

CLOSING THE LOOP: VALUE GAP IDENTIFICATION

The UX doesn't just deliver value — it reveals where value is missing.

UX Signal	What It Means	Gap Identified	Action Triggered
Null Result Patterns	Queries returning zero or low-confidence results	Content gap — knowledge exists but isn't ingested	Content acquisition or ingestion priority
Escalation Clusters	Queries consistently forwarded to human experts	Agent capability gap — topic not covered by any agent	New domain agent training or knowledge curation
Low Acceptance Rate	Results returned but not engaged with	Relevance gap — semantic matching needs tuning	Persona ontology or relevance model refinement
Repeated Query Loops	Same user, same topic, multiple sessions	Learning gap — knowledge isn't being retained or applied	Learning pathway recommendation triggered
Cross-Persona Divergence	Same query, dramatically different results for different roles	Vocabulary alignment gap in persona ontology	Ontology governance review flagged
Innovation Query Signals	Novel queries with no ontology coverage	Strategic gap — emerging knowledge domain not yet mapped	Knowledge engineering backlog item created

THE CONTINUOUS IMPROVEMENT LOOP



The UX abstraction boundary ensures this loop runs continuously without disrupting the employee experience — improvement is invisible, impact is visible.

IMPLEMENTATION ROADMAP

Phase 1

Months 1–4

Semantic Foundation

- ▶ ARCH ontology extensions for CoLE domain
- ▶ Content connector scaffolding (Library, DMS, LMS)
- ▶ Persona ontology: first 3 role families
- ▶ NLP→SPARQL prototype & validation

Phase 2

Months 5–9

Agent Build-Out

- ▶ Scientific & Compliance agents deployed
- ▶ Learning Pathways agent live
- ▶ Full-text ingestion: PubMed + internal white papers
- ▶ UX beta with metrics capture framework

Phase 3

Months 10–14

CoLE Integration

- ▶ CoLE UX general availability
- ▶ Value dashboard for CoLE leadership
- ▶ Value gap detection loop operational
- ▶ All 4 content domains federated

Phase 4

Months 15–18

Intelligence Expansion

- ▶ Adaptive learning path recommendations
- ▶ Expert network discovery layer
- ▶ Innovation signal detection
- ▶ Cross-site global vocabulary harmonization

ARCH semantic infrastructure collapses Phase 1 from a 12-month build to a 4-month extension — the foundation already exists.

WHY THIS WINS

Most enterprise learning platforms compete on content breadth. This competes on knowledge precision.

100%

Cited Responses

Semantically Grounded

Not RAG with vector similarity — ontology-guided retrieval that respects the relationships between knowledge objects. Every response is traceable to a source. Hallucination is architecturally constrained, not prompted away.

Day 1

Zero Training

Genuinely Personalized

The persona model translates vocabulary invisibly. An MSL and a regulatory affairs specialist asking the same question get different, role-appropriate answers — without any configuration by the user.

Continuous

Value Loop

Measurably Valuable

The UX is a measurement instrument. Every interaction generates value signal. CoLE leadership sees knowledge gap maps, content ROI, and learning impact — not just usage counts.

The ARCH foundation exists. The semantic infrastructure is built. This is an extension, not a greenfield project.

PROPOSED NEXT STEPS

01

CoLE Discovery Workshop

Map the 3 highest-priority persona families and their top 5 jobs-to-be-done. Define value measurement success criteria with CoLE leadership.

Week 1–2

02

System Inventory & API Mapping

Document all target content systems. Assess API availability, data quality, and semantic readiness for ingestion pipeline design.

Week 2–4

03

ARCH Ontology Gap Analysis

Evaluate existing ontology coverage against CoLE knowledge domains. Identify extension points and new modules required.

Week 3–5

04

UX Prototype & Metrics Design

Build conversational interface prototype with metrics capture framework. Validate persona vocabulary mapping with representative users.

Month 2–3

05

Pilot Program — 2 Persona Families

Deploy limited pilot with Scientific and Regulatory personas. Capture value signal data. Demonstrate gap identification loop in action.

Month 3–5